

ASSIGNMENT - VI

VI. Problems based on properties of Laplace Transforms :

1. Find $L^{-1} \left[\log \left(\frac{s+a}{s-a} \right) \right]$

Ans : $2 \left[\frac{\operatorname{atcosh} at - \operatorname{sinh} at}{t^2} \right]$

2. Find $L^{-1} \left[\cot^{-1} \left(\frac{s+a}{b} \right) \right]$

Ans : $\frac{e^{-at} \sin bt}{t}$

3. Find $L^{-1} \left[\frac{1}{(s^2+a^2)^2} \right]$

Ans : $f(t) = \frac{\sin at - at \cos at}{2a^3}$

4. Find $L^{-1} \left[\frac{1}{s^4 - 2s^3} \right]$

Ans : $f(t) = \frac{e^{2t} - 2t^2 - 2t - 1}{8}$

5. Find $L^{-1} \left[\frac{e^{-3s}}{s^2 - 2s + 5} \right]$

Ans : $f(t) = \frac{e^{(t-3)} \sin 2(t-3) u(t-3)}{2}$